

The Evolution of Cooperation

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A Question from Last Time

The Backward Induction Argument

Last time we left with a puzzle.

In a repeated Prisoner's Dilemma with a **known endpoint**, backward induction says you should defect in the last round, because there's no future to protect.

But if you'll defect in the last round, your opponent knows that. So they'll defect in the second-to-last round. And so on, all the way back to round one.

Does this mean rational players can *never* cooperate?

Why It Doesn't (Quite) Work That Way

Technically, the argument is valid for a **finite, known-length game** between perfectly rational players who know each other are perfectly rational.

But three things complicate this:

- Players don't actually reason that way in practice
- If an opponent cooperates early, you have good reason to cooperate back
- If the **endpoint is uncertain**, the unraveling argument fails entirely

The more interesting question: can we explain *why* cooperation emerges, and what strategies actually work?

Axelrod's Experiment

In 1980, Robert Axelrod, a political scientist here at the University of Michigan, ran an unusual experiment.

He **invited researchers** from economics, psychology, mathematics, sociology, and computer science to submit strategies for playing the iterated Prisoner's Dilemma.

Each strategy was a computer program. Axelrod ran a **round-robin tournament**: every strategy against every other strategy, repeated many times. Whoever accumulated the most points overall won.

The First Tournament

The first tournament had **14 entries**. The strategies ranged from simple to highly sophisticated. Some tried to detect and exploit patterns, some started with defection, some tried complex conditional logic.

The winner was the **simplest program submitted**: four lines of code.

It was submitted by the game theorist Anatol Rapoport, and it was **Tit for Tat**: cooperate on the first move, then copy whatever your opponent did last round.

The Second Tournament

Axelrod published the results and ran a **second tournament**. Now **62 entries** came in.

People knew TFT had won. Some tried to beat it directly. Some submitted variants. Some entered highly sophisticated strategies designed to detect and outmaneuver it.

TFT won again.

Axelrod published *The Evolution of Cooperation* in 1984. It became one of the most widely cited books in the social sciences. It's become very important in **biology**, as well as across PPE.

- Each pair of strategies played **200 rounds** of the Prisoner's Dilemma
- Every strategy faced every other strategy (round-robin)
- Total score = points accumulated across all matches

How the Tournament Worked

The payoff matrix Axelrod used:

	Cooperate	Defect
Cooperate	3, 3	0, 5
Defect	5, 0	1, 1

Mutual cooperation earns 3 each. Mutual defection earns only 1 each. The temptation to defect is real, but exploitation backfires if the other player responds.

Note one key feature of this payoff structure. Alternating CD/DC does **worse** than continuous CC. Some people make this definitive of the game being a PD.

Our Tournament

Basically the same setup as Axelrod's.

But instead of submitting code, you'll choose from a menu of pre-built strategies.

You'll answer a short questionnaire:

1. Are you a **Reciprocator** (react to what your opponent does) or a **Pattern Hunter** (probe and exploit)?
2. Whatever answer you give, you'll get a question asking for more detail about the kind of view you have.
3. Then you'll pick from the 2–3 strategies in your category

Then I'll run the round-robin tournament and we'll see what happens.

Reciprocators react to what their opponent does. Here are some prominent examples:

- **Tit for Tat:** Cooperate first, then copy whatever the opponent did last round.
- **Generous TFT:** Like TFT, but randomly forgives about one in ten defections.
- **Grim Trigger:** Cooperates until the first defection, then defects forever.

Pattern Hunters probe and exploit. For example:

- **Cooperator Hunter:** Tests whether the opponent has ever defected and if not, exploits them forever.
- **Prober:** Opens with D, C, C to test for exploitability, then defects forever if the probe succeeds.
- **Cycle Hunter:** Watches for any repeating pattern in the opponent's

You'll be asked: **How forgiving are you?**

- **Very forgiving:** Occasional defections don't bother you. → *Tit for Two Tats, Generous TFT*
- **Standard:** Mirror what your opponent does. → *Tit for Tat, Omega TFT*
- **Strict:** Punish defection firmly and take a while to let it go. → *Two Tits for Tat, Hard TFT, Soft Grudger*
- **Unforgiving:** Once betrayed, you never cooperate again. → *Grim Trigger, Spiteful TFT*

You'll be asked: **What are you hunting for?**

- **Exploitable cooperators:** Find opponents who won't fight back, and safely defect on them forever. → *Cooperator Hunter, Prober, Hard Prober*
- **Predictable patterns:** Detect opponents playing a fixed, mechanical strategy and outmaneuver them. → *Alternator Hunter, Cycle Hunter, Defector Hunter*



Figure 1: QR code for IPD Game

Go to:

bweatherson.github.io/games/ipd-game.html

Answer the questionnaire, pick your strategy, and enter your name.

Once everyone has joined, I'll run the tournament and results will appear on your screen.

(Tournament results here)

What Makes Strategies Succeed

After the tournaments, Axelrod analyzed what separated the successful strategies from the unsuccessful ones.

He found that success didn't come from being clever, aggressive, or unpredictable. It came from four surprisingly simple properties, plus a lesson about what counts as success.

Never be the first to defect.

Of the top eight strategies in the second tournament, *all eight* were “nice”.
They were never the first to defect.

None of the strategies that defected first finished in the top half.

This seems counterintuitive. Shouldn't you gain an edge by defecting first?

A nasty strategy can do well **once**, but it then plays *the rest of the game* in a damaged state.

Once you've defected first, your opponent retaliates. Now you're both locked into defecting, earning 1 per round instead of 3.

A nice strategy avoids this entirely. Against another nice strategy, it earns 3 per round for all 200 rounds. That adds up.

Let's say we ran the same tournament, but didn't add up points. Instead we treated everything like a basketball game, and asked who *won* each pairing.

What's the best strategy to play in that game?

It's **All D**. It wins every time. But it's often the **worst** strategy using the rules as we've defined them.

This matters for thinking about what real world situations can be modeled this way.

Respond to defection quickly.

Being nice doesn't mean being a pushover. Strategies that *never* defected (always cooperate) got exploited by anything that defected repeatedly, earning 0 per round instead of 3.

TFT retaliates *immediately*: defect on me this round, I defect back next round. No delay, no forgiveness yet.

This makes exploitation unprofitable from the start.

Return to cooperation after you've punished.

Strategies that retaliated but never forgave got locked into **mutual defection loops**. They earn 1 per round for the rest of the match. That's almost as bad as being exploited.

TFT forgives immediately: as soon as the opponent cooperates, TFT cooperates on the next move.

Quick retaliation *plus* quick forgiveness keeps punishment short and gets both players back to earning 3 per round.

Make your strategy legible to your opponent.

TFT is easy to understand: cooperate first, then mirror whatever your opponent just did. Any opponent can figure this out within a few rounds.

Many sophisticated strategies were essentially unreadable. Opponents couldn't tell if they were dealing with something cooperative or exploitative.

When opponents can't figure you out, they defect defensively, and you lose points you could have made through cooperation.

Axelrod noticed that many people framed the tournament wrong: they tried to **beat** their opponents. Taking that to an extreme, you might play **All D**.

But **TFT** *never* beats any individual opponent. Against a defector it breaks even; against a cooperator it ties; it only “loses” when the opponent defects first.

The goal isn't to beat your opponent. It's to do well across all your matches.

A strategy that earns 3 per round against many opponents accumulates more points than one that “wins” against a few while earning 1 per round against everyone else.

What Happened in Our Tournament?

Looking at our results:

- Which strategies finished at the top? Were they nice?
- Did any hunters outperform the reciprocators overall?
- What happened when two hunters played *each other*?
- Would the result change if the field had more of one type?

Cooperation can emerge from **self-interest alone**: without altruism, central authority, or communication.

All that's required:

- The interaction repeats (there's a “shadow of the future”)
- Players can recognize each other's behavior and respond to it
- A strategy that rewards cooperation and punishes defection is available

Axelrod showed this works even in **evolutionary** settings: if successful strategies spread and unsuccessful ones die out, cooperative strategies dominate over time, even starting from a world of defectors.

Axelrod's work raises as many questions as it answers:

- Does **TFT** always win, or does it depend on who else is in the field?
- What happens to cooperation in large populations over time?
- How does this apply beyond game theory, to arms races, trade, and everyday trust?